

Project Report - Team 4

Seek, Reason, Resolve: An Agentic Approach to Intelligent Question Answering

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May 18, 2026

The relevant code for this paper can be found [here](#).

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1 Application Area and Goals

This project addresses web-based factual question answering with retrieval-augmented generation (RAG) and LLM-based agents. The main motivation is that large language models can produce fluent but unsupported answers, especially for fact-focused questions that require exact entities, dates, or relations. To reduce this problem, our system combines live web retrieval, evidence selection, and agentic control steps. Instead of answering directly from model memory, it retrieves current web evidence, filters useful information, and generates a grounded answer based on that evidence.

In other words, the core problem addressed in this project is factual question answering over live web data. Given a natural-language question, the system retrieves evidence from the open web, selects the most relevant snippets, and generates a final answer that is strictly supported by the retrieved material. The primary objective is not merely to produce a correct answer, but to ensure that every claim is traceable to specific evidence. In practical terms, the task integrates six interconnected stages: user question, planning, information retrieval, evidence control, answer and critique generation, and final answer.

To achieve this, the project aims to build a functioning web-based Q&A system that leverages Tavily for live retrieval and OpenRouter-backed large language models for generation and reasoning. Beyond establishing a baseline pipeline, the work seeks to develop an advanced multi-agent Retrieval-Augmented Generation (RAG) architecture featuring configurable retrieval and reasoning components. This modular design allows for a systematic evaluation of different architectural variants to determine their impact on performance. Ultimately, the project intends to analyze the trade-offs between answer quality, grounding fidelity, citation reliability, and computational efficiency, moving beyond simple accuracy metrics to understand the costs and benefits of adding agentic complexity to the retrieval process.

2 Data and Task Profile

The project evaluates a factual question answering system in a live web-retrieval setting. This means the system does not answer from a fixed internal corpus, but retrieves evidence dynamically from the open web during runtime. As a result, the task is more realistic than static-corpus QA, but also more difficult because retrieved information can be noisy, incomplete, duplicated, or unstable across runs.

2.1 Datasets and Evaluation Scope

The implemented evaluation utilizes three primary factual Q&A benchmark datasets: **SimpleQA** (OpenAI 2024b), **PopQA** (Mallen et al. 2022), and **TriviaQA** (Joshi et al. 2017). These datasets were selected because they represent distinct types of factual questions, thereby testing different aspects of the system’s capabilities. Specifically,

SimpleQA focuses on short, precise factual questions requiring compact answers, making it particularly useful for testing *exactness* since even minor deviations can render an answer incorrect. **PopQA** contains questions centered around entities and relations, serving to test whether retrieval and verification mechanisms improve reliability beyond the model’s likely memorized knowledge. Finally, **TriviaQA** consists of broader and often more challenging factoid questions, which are used to evaluate *retrieval robustness* and to determine if stronger ranking and evidence selection strategies significantly improve performance on complex queries.

While these three datasets form the core of our evaluation, we also considered *BrowseComp* (OpenAI 2024a) due to its status as a standard benchmark for high-tier agents. However, we excluded it because the dataset requires operating an actual web browser rather than a search engine. The tasks often involve visual reasoning (e.g., identifying buildings from images) or spatial navigation (e.g., using maps to locate nearby structures), which exceeds the scope of our text-based retrieval pipeline.

The nature of these datasets and the live web setting introduces specific challenges. This QA setting demands exact factual precision despite the inherent noise, incompleteness, and potential conflicts found in search results. Consequently, the system must not only identify relevant evidence but rigorously verify its sufficiency and maintain clear citation grounding when synthesizing multiple sources. To address these challenges and ensure a balanced evaluation, we constructed a custom dataset comprising **50 questions** with a deliberate distribution: 30 from SimpleQA, 10 from TriviaQA, and 10 from PopQA. We allocated the majority of questions to SimpleQA because it effectively serves as a predecessor to BrowseComp, offering similarly challenging factual reasoning tasks without the restrictive requirement of operating an actual web browser or handling multimodal inputs. This allows us to test the limits of our text-based retrieval pipeline on complex reasoning while avoiding the architectural bottlenecks of the full BrowseComp benchmark.

2.2 Data Structure and Output Format

The evaluation pipeline uses structured benchmark and prediction records so that each system output can be traced back to the original question and to the configuration that produced it. At the input side, benchmark instances typically contain a question ID, the question text, the dataset name, and a gold answer. At the output side, the system stores the selected setup, the generated answer, the retrieved and selected evidence, the critic result, and evaluation-related metadata such as source count, evidence count, and latency.

This structured representation is crucial for reproducible evaluation. It allows predictions from different system variants to be compared systematically across datasets and metrics. Furthermore, it supports later error analysis by linking answer quality to the specific evidence available during generation and the architectural components enabled in a given run.

3 System Architecture and Workflow

The system is designed as an orchestrated Retrieval-Augmented Generation (RAG) workflow for web-based factual question answering. A central orchestrator manages the control flow and coordinates dedicated components for query planning, retrieval, evidence processing, answer generation, and critique. Several of these components are LLM-based agents, including the query planner, HyDE generator, evidence filter, evidence sufficiency, answer writer, and critic. Other parts of the system, such as web search, chunking, sparse and dense retrieval, hybrid score fusion, and cross-encoder reranking, are implemented as retrieval and ranking modules. This makes the design close to an *agent harness*: LLMs perform task-specific reasoning within controlled components, while the surrounding software manages tool access, state, retrieval iterations, evidence merging, and logging. All advanced components can be individually enabled or disabled through configuration flags, which supports the ablation comparisons described in 3.3.

3.1 Baseline Pipeline

As a reference point for the experimental comparison, we define a lightweight single-pass RAG baseline. The baseline follows a simple retrieval-generation structure:

question → web search → chunking → lexical retrieval → answer writer → critic.

The original user question is sent directly to Tavily web search, after which the retrieved documents are split into overlapping text chunks. Candidate chunks are ranked with a lexical baseline that scores each chunk by the number of overlapping tokens with the user question. The Answer Writer Agent then generates a response from the selected evidence, while the Critic Agent evaluates whether the answer appears grounded and relevant. In the baseline configuration, this critique is recorded but does not trigger retrieval retries or answer revision. Query decomposition, HyDE, hybrid retrieval, cross-encoder reranking, evidence filtering, evidence sufficiency checks, iterative retrieval, and Self-RAG-style reflection are all disabled. The baseline thereby provides a simple comparison point for assessing the contribution of the more advanced components.

3.2 Advanced Multi-Agent RAG Pipeline

The advanced pipeline extends the baseline with additional planning, retrieval, and evidence-control components. Figure 1 illustrates the overall architecture. The process starts with a natural-language user question and proceeds through five further stages: planning, retrieval and ranking, evidence control, answering and critique, and final answer construction.

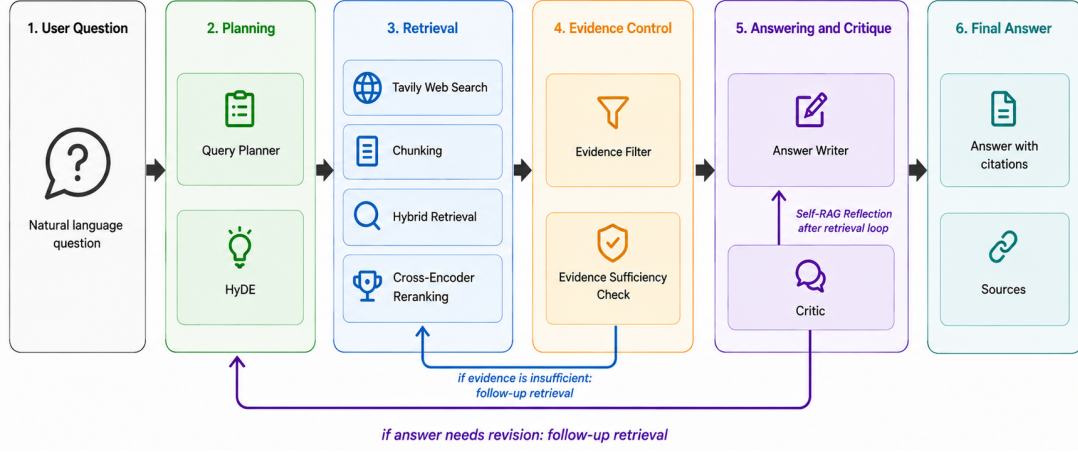


Figure 1: Architecture of the multi-agent RAG question answering system.

In the planning stage, the Query Planner Agent processes the original user question and generates one or more focused web-search queries together with a Tavily search depth setting. This decomposition is particularly useful for complex or ambiguous questions where a single query may not retrieve sufficient evidence. Independently, the HyDE Agent generates a hypothetical evidence passage directly from the original user question. This passage is not treated as factual evidence and is never cited in the final answer. Instead, it is appended to the original question to form the retrieval text used during sparse and dense ranking, reducing the semantic gap between short questions and longer web documents.

The retrieval stage collects and ranks evidence candidates. Tavily web search retrieves results for the planned queries, providing title, URL, and page content for each result. The retrieved text is split into overlapping chunks, which become the candidate evidence units. In the advanced setup, the system applies hybrid retrieval by combining sparse BM25-based lexical retrieval with dense embedding-based semantic retrieval. Sparse retrieval is well-suited for exact names, dates, and keywords, while dense retrieval identifies semantically related passages even when the wording differs from the question. The implementation combines normalized sparse scores, normalized dense scores, and a Reciprocal Rank Fusion (RRF) signal into a single hybrid score. The top candidates are then passed to a cross-encoder reranker, which jointly scores the original question and each candidate passage to further improve precision.

After retrieval and reranking, the system applies several evidence-control steps before answer generation. Evidence from the current and any previous retrieval rounds is first merged and deduplicated by URL and text. The Evidence Filter Agent then uses an LLM to select the most relevant evidence chunks, retaining passages that are directly useful for answering the question and discarding irrelevant or redundant ones. After filtering, evidence items are renumbered into a clean sequence (E1, E2, E3, ...) to maintain citation consistency for both the answer writer and the critic.

The Evidence Sufficiency Agent subsequently evaluates whether the retained evidence is sufficient to support a grounded answer. If relevant information appears to be missing, it returns the missing aspects together with focused follow-up search queries. The orchestrator can then use these queries to trigger additional retrieval retries before answer generation. This mechanism operates within the current pipeline iteration and is separate from the later critic-driven retrieval loop.

The Answer Writer Agent generates a response using only the selected evidence and is prompted to cite evidence IDs for factual claims. The Critic Agent then receives the generated answer together with the same evidence pool and evaluates groundedness and relevance, returning structured feedback that includes a binary revision signal and a free-text comment. The Critic Agent does not verify whether each individual citation supports the specific claim it is attached to. Citation-level support is assessed separately in the evaluation pipeline.

If the Critic Agent indicates that revision is needed and further retrieval iterations are available, the orchestrator can trigger another retrieval loop. The Query Planner Agent generates new search queries based on the original question, the current answer, and the critic comment. The system then retrieves and merges additional evidence before producing a revised answer. Once the retrieval loop has concluded, the system can additionally apply a Self-RAG-inspired reflection step if the Critic Agent still requests revision. Unlike iterative retrieval, this step does not search for new evidence, but prompts the Answer Writer Agent to revise the answer using the final selected evidence and the critic feedback. These mechanisms are designed to help the advanced pipeline recover from incomplete initial retrieval and improve answer grounding. Their actual impact on answer quality is assessed empirically in the evaluation.

3.3 Experimental Variants and Ablation Study

The incorporation of all those components raises the need of an ablation study to analyze how individual components affect the performance of our system. The system was designed with several configurable components because Q&A multi-agent RAG systems can benefit from different retrieval, filtering, and reflection mechanisms depending on the task. The ablation study therefore investigates whether the additional architectural complexity improves performance compared to a simpler baseline. In each ablation run, one optional component was disabled while the remaining components were kept active. This leave-one-out design was chosen due to time constraints; evaluating all possible combinations of the eight optional components would require $2^8 = 256$ experiments.

The eight components considered in the ablation study were: *query decomposition*, *hybrid retrieval*, *cross-encoder reranking*, *iterative retrieval*, *evidence filtering*, *evidence sufficiency checking*, *HyDE*, and *Self-RAG-style reflection*. In addition to the eight ablation runs, we evaluated two reference configurations: a baseline system without any of these optional components and a full system with all components enabled. In total, this resulted in ten experimental variants.

By removing one component at a time and comparing the resulting performance to the

full system, we can identify which modules mainly contribute to retrieval quality, answer grounding, or runtime efficiency. Since components interact with each other, these results should be interpreted as indicative rather than strictly causal. The ablation study therefore not only compares the full system against the baseline, but also shows which parts of the architecture are most relevant for answering different types of questions. The actual impact of these components is analyzed in 5.1.

4 Evaluation Setup

4.1 Evaluation Metrics

Metric definitions Let $\mathcal{Q}$ denote the question, $\mathcal{A}$ the generated answer, and $\mathcal{S}_{\mathcal{A}}$ the set of all sources cited in $\mathcal{A}$. The metrics are defined as follows:

$$\text{Correctness } \mathcal{C} = \begin{cases} 1 & \text{if } \mathcal{A} \text{ is correct} \\ 0 & \text{if } \mathcal{A} \text{ is rejected} \\ -1 & \text{if } \mathcal{A} \text{ is wrong} \end{cases} \quad (1)$$

$$\text{Relevance } \mathcal{R} = \begin{cases} 1 & \text{if } \mathcal{A} \text{ is relevant to } \mathcal{Q} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Let c_s denote the specific claim in answer $\mathcal{A}$ to which source $s \in \mathcal{S}_{\mathcal{A}}$ is attached. We define the per-source support indicator $\mathbb{I}_{\mathcal{A}}(s)$ as:

$$\mathbb{I}_{\mathcal{A}}(s) = \begin{cases} 1 & \text{if } s \text{ supports its corresponding claim } c_s \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Citation accuracy is then the arithmetic mean of these source scores:

$$\text{Citation Accuracy} = \frac{1}{|\mathcal{S}_{\mathcal{A}}|} \sum_{s \in \mathcal{S}_{\mathcal{A}}} \mathbb{I}_{\mathcal{A}}(s) \quad (4)$$

Evaluation is performed via LLM-as-a-judge. The judge estimates **Correctness** by comparing the answer to the gold and assigning a score per Equation 1. **Answer Relevance** is binary and indicates whether the answer is on topic, regardless of correctness Equation 2. **Citation Accuracy** measures whether the evidence in each cited source $s \in \mathcal{S}_{\mathcal{A}}$ actually supports the corresponding claim.

For finer-grained estimation we additionally report RAGAS metrics. **Faithfulness** measures whether the answer is grounded in the retrieved sources. **RAGAS Answer Relevance** serves the same purpose as ours but is computed via embeddings. **Context Recall & Precision** are gold-conditioned: Recall measures whether the gold facts are

present in the retrieved context, and Precision whether the relevant contexts are ranked high.

Other metrics suggested in the project outline, F1, BERTScore, Recall@k, Precision@k, NDCG, were avoided, since they require either a fixed reference answer for token- or embedding-level matching, or a static gold ranking of retrieved sources, neither of which fits dynamically retrieved web content¹.

System efficiency during ablation runs is reported via **Latency** only. The total cost, including 1-2 incorrect runs, is approximately 41\$, which is much lower than the prior manually estimated cost (≈ 1.5 M tokens per setup) due to automatic token caching from the model providers.

4.2 Evaluation Workflow

The evaluation workflow is split into two phases: **Predicting** and **Scoring**. In the prediction phase the system answers each benchmark question using either the full pipeline or the specific ablation under test. The scoring phase then evaluates those predictions with the metrics above. Results from both phases are stored in JSONL.

5 Results, Error Analysis, and Discussion

5.1 Results

Across the baseline, full, and per-feature ablation runs, the majority of answers were correct. Each setup also produced a small fraction of rejected (no-answer) responses and a slightly larger fraction of wrong responses [Figure 2](#). The deviations in correctness rate between setups are small, and we observe anomalies such as the full and baseline setups landing on the same correctness rate. This is largely an artifact of the small benchmark size: with 50 questions per run, a single mishap from the agent or an unlucky web search shifts correctness by 2 percentage points.

Answer Relevance is very consistent across setups, sitting at 100% throughout, with a single 2-point dip in the run that excludes the Cross-Encoder Reranking agent [Figure 3](#). RAGAS Answer Relevance is similarly stable but lower, ranging from 82% to 90%, reflecting its stricter embedding-based comparison [Figure 3](#).

Citation Accuracy also remains consistently above 80%, as each factoid answer can typically be anchored to an explicit supporting passage within its cited source [Figure 3](#).

Context Recall is tight at 72%-86%, meaning the gold facts are reliably present somewhere in the retrieved context across all setups. Context Precision varies more widely, from 58% to 88%: the baseline (58%) and the w/o filter run (62%) sit clearly below the rest, indicating that the Evidence Filtering agent boosts precision substantially without hurting recall [Figure 3](#).

¹The $*@k$ metrics were excluded together with the Meta CRAG Benchmark, as the dynamic nature of the web makes URL-based static evaluation of retrieved sources practically infeasible.

Faithfulness is the lowest-valued metric overall, with all setups falling between 47% and 61%. The baseline (which performs no evidence filtering) sits highest at 61%, suggesting that the aggressive context pruning performed by the filtering agent may leave fewer passages against which RAGAS can verify each claim [Figure 3](#).

Latency differences between the full run, the per-feature ablations, and the baseline show the baseline is by far the fastest because it skips every additional agent. Among ablations, removing the Evidence Filtering agent reduced response latency, while removing the Hybrid Retrieval agent increased it.

Removing the Evidence Filtering agent skips one LLM stage and also reduces the chance that the Evidence Sufficiency agent triggers retry searches on a too-small filtered set, both of which plausibly account for the latency drop. The latency increase from removing the Hybrid Retrieval agent runs in the opposite direction and is best read alongside the noise caveat below.

The latency picture should be read with caution: our runs were spread unevenly across the week and are therefore heavily influenced by external noise such as fluctuating token throughput from the LLM provider.

5.2 Error Analysis

We first group the 50 benchmark questions by how many of the ten setups answered each correctly (see [Table 1](#)). More than half (26 questions) sit at the ceiling, meaning every setup answers them correctly, and three sit at the floor where no setup did. Only the remaining 21 questions actually discriminate between setups, and they are dominated by SimpleQA (14) with smaller contributions from PopQA (4) and TriviaQA (3). All observed differences in mean correctness across setups are therefore driven by this 21-question subset.

Group	PopQA	SimpleQA	TriviaQA	Total
Ceiling (10/10)	3	16	7	26
Discriminating (1-9)	4	14	3	21
Floor (0/10)	3	0	0	3
Total	10	30	10	50

Table 1: Distribution of the 50 benchmark questions by the number of setups (out of 10) that answered them correctly.

5.3 Discussion

A complementary human evaluation of eight stratified samples revealed a critical divergence in automated assessment reliability. While the LLM judge achieved perfect alignment with human raters on correctness and answer relevance (100% exact match),

it exhibited severe instability on citation accuracy, matching human scores in only 37.5% of cases. Quantitative analysis shows negligible correlation ($r = 0.17$) and a substantial Mean Absolute Error of 0.36, indicating an average deviation of over one-third of the scoring scale. Notably, in two SimpleQA instances, the judge assigned perfect scores (1.00) to citations human raters deemed entirely unsupported (0.00), suggesting a tendency to hallucinate grounding where evidence was truncated or absent. This statistical gap confirms that while the judge reliably assesses factual correctness, its evaluation of evidentiary grounding is highly inconsistent.

The differences in correctness observed across our ten setups are small in absolute terms and fall well within the sampling noise expected at $n = 50$ questions per run, where each individual prediction shifts the mean by two percentage points. As shown in Table 1, only 21 of the 50 questions actually discriminate between setups; the remainder either sit at the ceiling (every setup correct) or the floor (no setup correct). The effective sample size for distinguishing one ablation from another is therefore much smaller than 50, and we cannot confidently attribute the observed deltas to any individual feature.

The two factors driving this are the simplicity of the questions in our mix and the small overall sample size. SimpleQA, PopQA, and TriviaQA consist largely of short factoid lookups that a single web search typically resolves, leaving little room for the additional retrieval and verification agents to differentiate themselves. Scaling the benchmark to several hundred questions and adding harder, multi-hop entries from BrowseComp would likely surface per-feature contributions that this mix cannot. We did not pursue that route here because of the project’s limited credit budget on OpenRouter and the limited time available for additional runs.

Despite the absence of a clear winning configuration, the cost-quality picture still admits a practical takeaway. The full pipeline takes roughly six times longer than the baseline (348s versus 60s mean latency) for the same 74% correctness rate, and the baseline is competitive or better on every reported metric except Context Precision, where the full pipeline reaches 81% against the baseline’s 58%. The full setup is therefore most defensible in deployments where the curated evidence list itself is a system output, rather than in settings that consume the final answer alone.

A further limitation is potential overlap between the benchmark questions and the generator’s pretraining data. SimpleQA, PopQA, and TriviaQA are long-standing public benchmarks built around Wikipedia and related sources, so some contamination is plausible. To the extent that the generator can recall answers from parametric memory rather than from the retrieved evidence, both absolute correctness scores and the discriminating power of any retrieval-side ablation are dampened, which is consistent with the 26-question ceiling we observe in Table 1.

5.4 State of the art

A like-for-like comparison to leaderboard state-of-the-art on these benchmarks would not be informative at $n = 50$ questions per setup, and our judge protocol differs from the official scoring used in each benchmark paper. We instead position the system relative

to the published techniques it composes: Self-RAG (Asai et al. 2024), HyDE (Gao et al. 2023), hybrid sparse and dense retrieval (Lin et al. 2021), and cross-encoder reranking (Nogueira and Cho 2019). The full pipeline is best understood as a re-implementation of these techniques applied to live web retrieval, and the contribution of this report is the ablation study itself rather than a new leaderboard result.

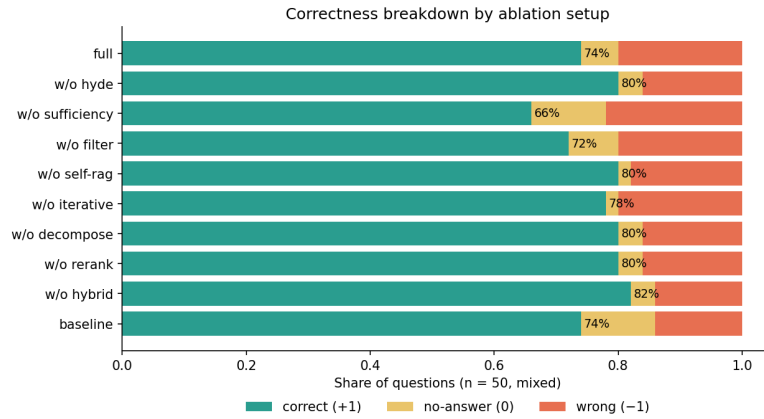


Figure 2: Correctness by setup breakdown



Figure 3: Metrics by setup

Table 2: Author Contributions

Subpart	I. Beck	P. Malamov	D. Maxheimer	R. Weber	V. Schmitt
Dataset selection	x		x		
Agent and module concept		x		x	
System architecture concept		x		x	
System architecture implementation				x	
RAG pipeline implementation				x	
Ablation study	x	x	x	x	x
Architecture review and feedback	x	x	x	x	x
Code review	x	x	x	x	x
Evaluation concept	x		x		
Evaluation harness	x		x		
Evaluation metrics	x		x		
Results analysis	x		x		
Human evaluation					x
Report writing	x	x	x	x	x
Review and editing	x	x	x	x	x

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Ehrenwörtliche Erklärung

Wir versichern, dass wir die beiliegende Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht haben. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Wir sind uns bewusst, dass eine falsche Erklärung rechtliche Folgen haben wird.

Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
GPT-5 & Codex	Rephrasing & Coding	Throughout	+

I. Beck *Ph. Lema* *Denis Marheine* *P. Weber* *Quix*

Mannheim, May 18, 2026